Healthcare Data Analysis Report (Excel Project)

1. Introduction

This project focuses on analyzing healthcare data using Microsoft Excel to identify patterns in patient health conditions, lifestyle factors, and medical costs. An interactive dashboard was created using pivot tables and charts to transform raw data into actionable insights.

2. Objective

- To analyze healthcare data using Excel
- To understand the impact of factors like smoking, weight, and diabetes
- To visualize healthcare trends through an interactive dashboard
- To support better healthcare decision-making

3. Dataset Description

The dataset includes the following key attributes:

- Smoker status (Yes/No)
- Diabetes status (Diabetes, Normal, Prediabetes)
- Weight status (Underweight, Normal, Overweight, Obesity)
- Hospital tier (Tier 1, Tier 2, Tier 3)
- BMI (Body Mass Index)
- Charges (Medical cost)
- HbA1c levels
- Number of children

4. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records
- Handled missing values
- Standardized categorical values (e.g., Yes/No, Weight categories)

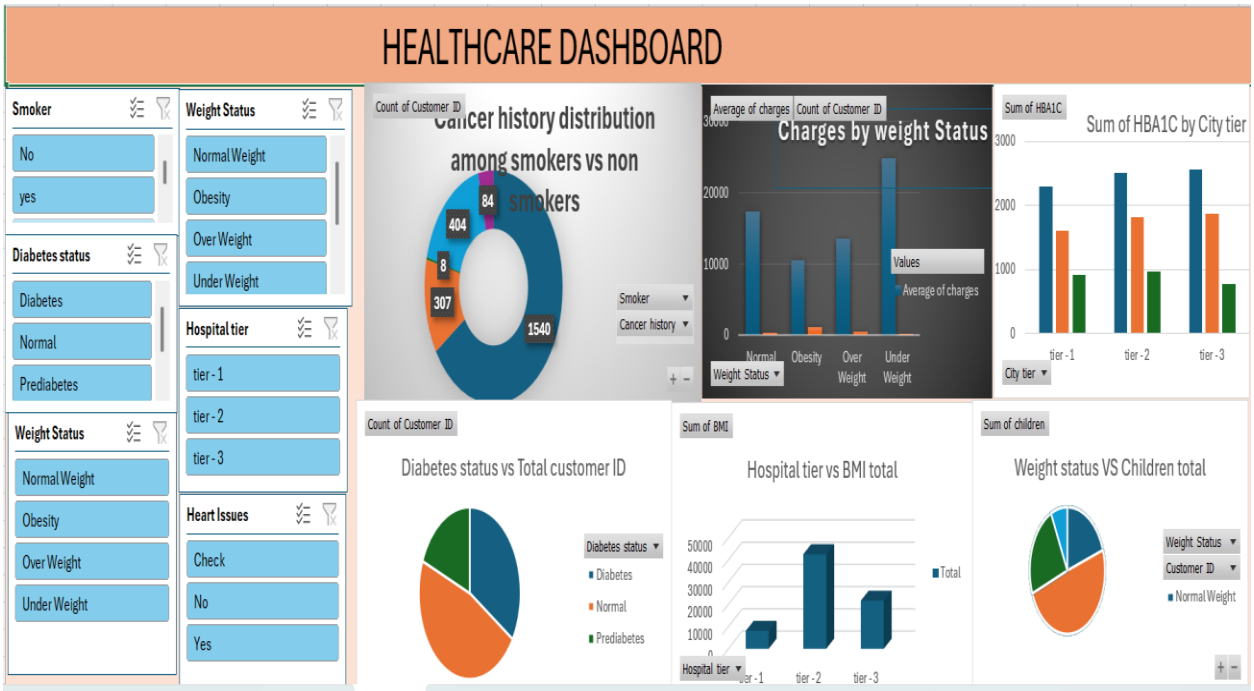
- Verified numerical columns (BMI, charges)
- Organized data for pivot table analysis

5. Tools & Techniques Used

- Microsoft Excel
- Pivot Tables
- Slicers (for filtering: smoker, weight, diabetes, hospital tier)
- Charts (Bar, Pie, Donut)
- Dashboard Design

6. Dashboard & Visualizations

An interactive dashboard was created with slicers and multiple charts for dynamic analysis.



Key Visuals:

- Cancer History (Smokers vs non-smokers)
  - Donut chart shows a significant difference between smokers and non-smokers
  - Smokers show higher counts in cancer-related data

- Charges by Weight Status
  - Underweight and obesity categories show higher medical charges
  - Normal weight individuals have relatively lower charges
- HbA1c by City Tier
  - Tier 2 and Tier 3 cities show higher HbA1c values
  - Indicates possible variation in healthcare quality or lifestyle
- Diabetes Distribution
  - Majority of individuals fall under Normal category
  - Significant portion in Prediabetes, indicating risk group
- Hospital Tier vs BMI
  - Tier 2 hospitals show higher total BMI values
  - Suggests higher patient volume or obesity prevalence
- Weight Status vs Number of Children
  - Overweight and normal weight groups have higher total children count
  - Possible demographic pattern

7. Key Insights

- Smokers are more associated with cancer-related conditions
- Medical charges increase with extreme weight conditions (underweight/obesity)
- Prediabetes population is significantly high → potential future risk
- HbA1c levels are higher in mid and lower-tier cities
- BMI trends indicate higher health risks in certain hospital tiers

8. Analysis Types

Descriptive Analysis

Summarized patient data using pivot tables and charts.

Diagnostic Analysis

Identified relationships between smoking, weight, and medical conditions.

Predictive Analysis

Observed trends indicating future health risks (e.g., prediabetes → diabetes).

Prescriptive Analysis

Suggested focus areas such as obesity control and early diagnosis.

9. Recommendations

- Promote awareness about smoking-related health risks
- Focus on weight management programs
- Early detection programs for prediabetes
- Improve healthcare access in Tier 2 and Tier 3 cities
- Monitor high-risk groups using dashboards

10. Learnings

- Building interactive dashboards using Excel
- Using slicers for dynamic filtering
- Data cleaning and structuring
- Converting raw data into insights
- Understanding healthcare analytics

11. Conclusion

This project demonstrates how Excel can be effectively used to analyze healthcare data and generate meaningful insights. The dashboard provides a clear view of how lifestyle factors like smoking and weight impact health outcomes and medical costs.

Overall, the analysis highlights the importance of preventive healthcare and data-driven decision-making in improving patient outcomes.